

5 INFLUENCE LINES

5.1 INTRODUCTION

So far we have studied the effects of applied loads acting at specified points on the structure. In certain structures the applied loads traverse the structure by moving from one end to the other. These moving loads are common in: -

1. Bridges – highways, railways, footbridges, etc
2. Industrial buildings – overhead cranes, conveyor belts, etc

The moving loads will have both **dynamic effects** and **static effects** on the structure. The dynamic effects are dealt with in FCE 401.

The static effects, namely support reaction components, axial force N , shear force V , and bending moment M , will vary with the position of the load. There is therefore a position at which a given moving load will produce the maximum effects on the structure.

The technique used to study the static effects on the structure as the load moves from one end of the structure to the other is based on the concept of an influence line

An influence line is a graphical representation of a specific load effect at a specific point due to a unit load traversing the given structure.

The unit load used here is real. It acts as the lowest factor for any other applied load.

5.2 PLOTTING INFLUENCE LINES

5.2.1 Using First Principles

Example 1

Plot the I.L.'s for V_A , V_B , V_C and M_C for the beam shown below.

Solution

Apply a unit load at distance x

(x measured from A)

I. L for V_A : $\sum M_B = V_A L - 1(L - x) = 0$

$$\Rightarrow V_A = 1 - x/L$$

I. L for V_B : $\sum M_A = 1(x) - V_B L = 0$

$$\Rightarrow V_B = x/L$$

I.L for V_C

For $0 \leq x \leq a$: $V_C = V_A - 1 = 1 - x/L - 1$

$$\Rightarrow V_C = -x/L$$

For $a \leq x \leq L$: $V_C = V_A = 1 - x/L$

$$\Rightarrow V_C = 1 - x/L$$

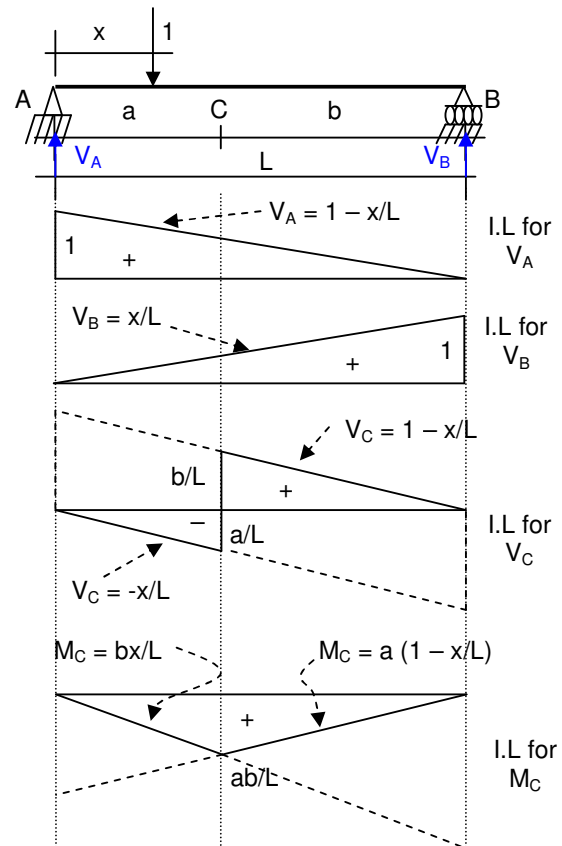
I.L for M_C

For $0 \leq x \leq a$: $M_C = V_A a - 1(a - x)$

$$\Rightarrow M_C = bx/L$$

For $a \leq x \leq L$: $M_C = V_A a = (1 - x/L)(a)$

$$\Rightarrow M_C = a(1 - x/L)$$



[NB: I.L.'s for determinate structures consist of straight lines. One only needs to determinate ordinates of a few control points, then connect these points by straight lines. The mathematical equations are NOT necessary].

Example 2

Plot the I.L.'s for V_A , V_B , V_C and M_C for the beam shown below.

Solution

Apply a unit load at distance x
(x measured from A)

I. L for V_A : $\Sigma M_B = V_A L - 1(L - x) = 0$

$\Rightarrow V_A = 1 - x/L$

I. L for V_B : $\Sigma M_A = 1(x) - V_B L = 0$

$\Rightarrow V_B = x/L$

I.L for V_C (shear force at C)

For $0 \leq x \leq L/2$: $V_C = V_A - 1 = 1 - x/L - 1$

$\Rightarrow V_C = -x/L$

For $L/2 \leq x \leq L$: $V_C = V_A = 1 - x/L$

$\Rightarrow V_C = 1 - x/L$

I.L for M_C (bending moment at C)

For $0 \leq x \leq L/2$

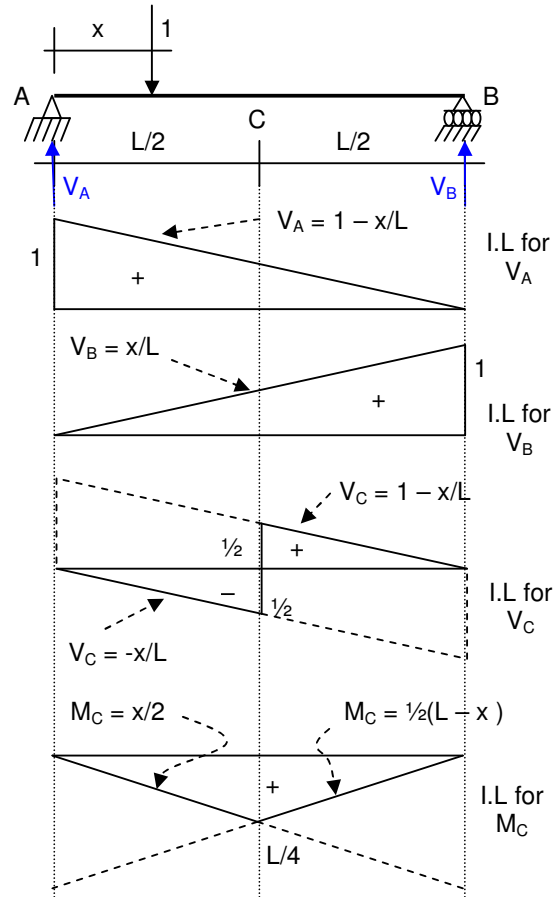
$M_C = V_A L/2 - 1(L/2 - x)$

$\Rightarrow M_C = x/2$

For $L/2 \leq x \leq L$

$M_C = V_A L/2 = (1 - x/L) (L/2)$

$\Rightarrow M_C = \frac{1}{2}(L - x)$



5.2.2 Using Second Principles

[Muller – Breslau Principle ©1890]

Muller-Breslau principle states that *the influence line for a specific load effect at a specific point is the same as the shape adopted by the structure when a unit displacement corresponding to the effect is imposed on the structure at that point, while keeping all other external forces unmoved*. [NB: The structure is considered a rigid body, therefore the adopted shape is linear, at least piece-wise].

Example 3

Plot the I.L.'s for V_A , V_B , V_C and M_C for the beam of example 1.

Solution

I. L for V_A

Give V_A unit displacement vertically upwards, keeping V_B unmoved. The resulting line is the **I.L for V_A**

I. L for V_B

Give V_B unit displacement vertically upwards, keeping V_A unmoved. The resulting line is the **I.L for V_B**

I.L for V_C

Imagine beam “cut” at C. Rotate AC and CB so that they end up being parallel (net rotation = zero), by applying a unit displacement as shown.

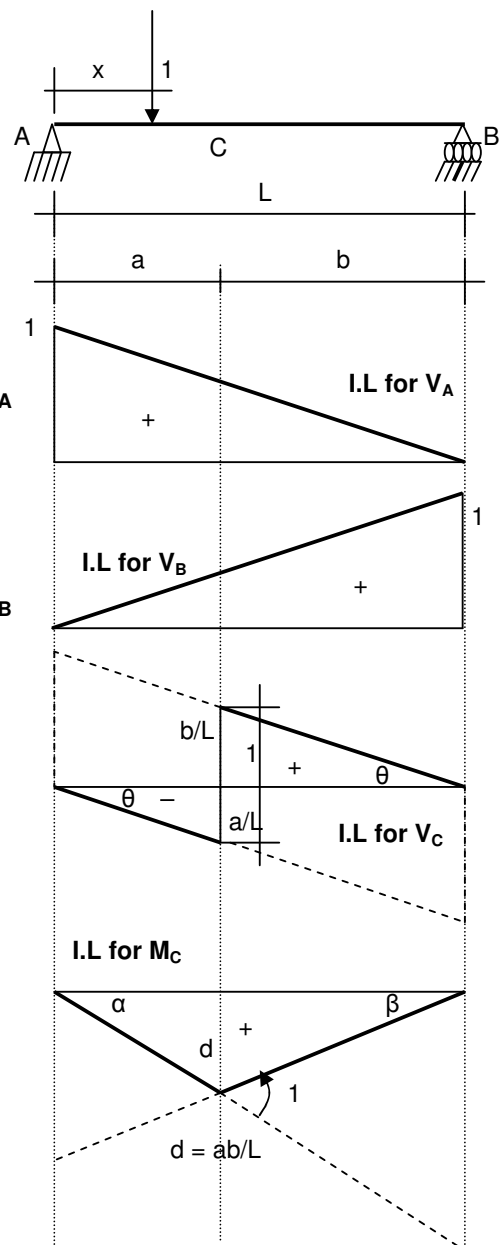
By compatibility $\theta a + \theta b = 1 \Rightarrow \theta = 1/L$

Hence obtain the ordinates a/L and b/L .

I.L for M_C

Imagine beam “fractured” at C. Give CB unit rotation relative to AC by displacing point C some distance d downwards. By compatibility: $\alpha + \beta = 1$

$$\Rightarrow d/a + d/b = 1 \Rightarrow d = ab/L$$



5.3 APPLICATION OF INFLUENCE LINES

Influence Lines are used to determine the **load effect at a specific point**, due to a **moving load at some other point**. This is achieved as follows:

1. Point load: Load effect (at a specific point) = Magnitude of load x Influence coefficient (at that other point).
2. U.d.l: Load effect (at a specific point) = Intensity of load x Area load covers under I.L.
3. Varying load: Load effect (at a specific point) = $\int$ (load function) x (I.L function) dx

Example 4

If $a = 6$ m and $b = 9$ m in example 3, determine the maximum V_A , V_B , V_C and M_C that

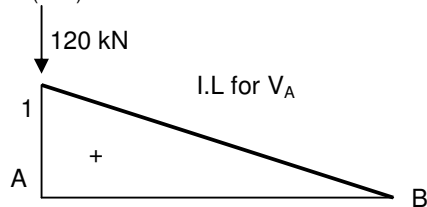
- a) A point load $P = 120$ kN
- b) A u.d.l. segment $\omega = 24$ kN/m of length 5 m

would induce at the points A, B and C while traversing the beam from A to B.

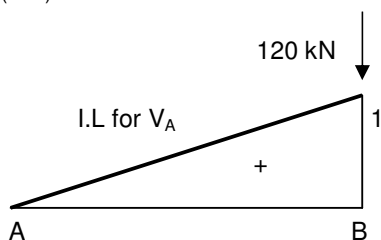
Solution

(a) Point load $P = 120$ kN

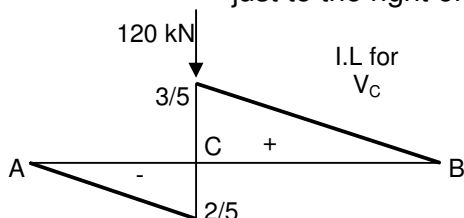
$$V_{A(\max)} = 120 \times 1 = 120 \text{ kN, } P \text{ at A}$$



$$V_{B(\max)} = 120 \times 1 = 120 \text{ kN, } P \text{ at B}$$

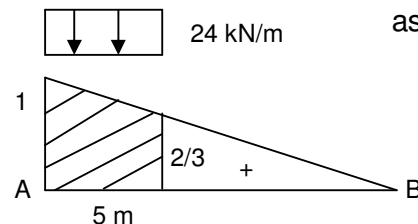


$$V_{C(\max)} = 120 \times (3/5) = 72 \text{ kN, } P \text{ just to the right of C}$$

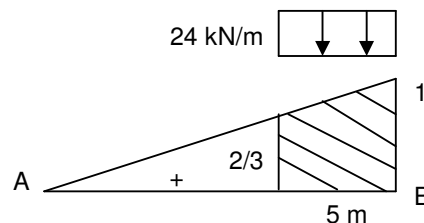


(b) U.d.l $\omega = 12$ kN/m

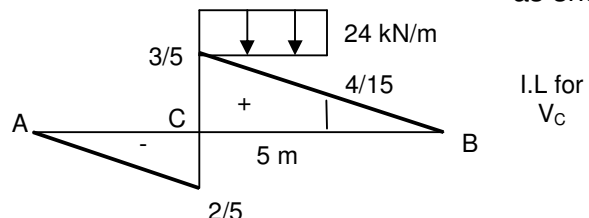
$$V_{A(\max)} = 24(1/2)(1+2/3)(5) = 100 \text{ kN, u.d.l arranged as shown.}$$



$$V_{B(\max)} = 24(1/2)(2/3+1)(5) = 100 \text{ kN, u.d.l arranged as shown.}$$

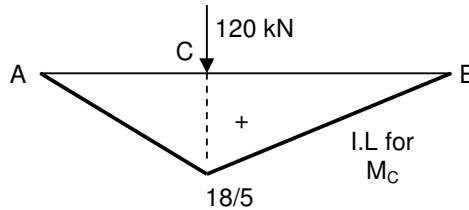


$$V_{C(\max)} = 24(1/2)(3/5+4/15)(5) = 52 \text{ kN, u.d.l arranged as shown}$$



$$M_{C \text{ (max)}} = 120 \times (18/5) = 432 \text{ kNm}$$

when P arrives at C



$M_{C \text{ (max)}}$: [The udl must be positioned strategically so as to cover maximum area under the I.L

How? See article 5.4

Note

In article 5.2.2 a simply supported beam was used to illustrate how to plot influence lines using second principles (Muller-Breslau principle). The influence lines for other beams may be plotted using the same technique. [See Tutorial # 6]

Question: What are the units of Influence coefficients?

Answer: It depends on the load effect being considered.

Homework: Complete the table below.

Diagram	AFD	SFD	BMD	M/EI	θ -diag	δ -diag	Influence Lines	
							Linear Force (Reaction or shear force)	Rotational Force (Bending Moment)
Units	kN	kN	kNm	m^{-1}	Dimensionless (Radians)	m	Dimensionless	m